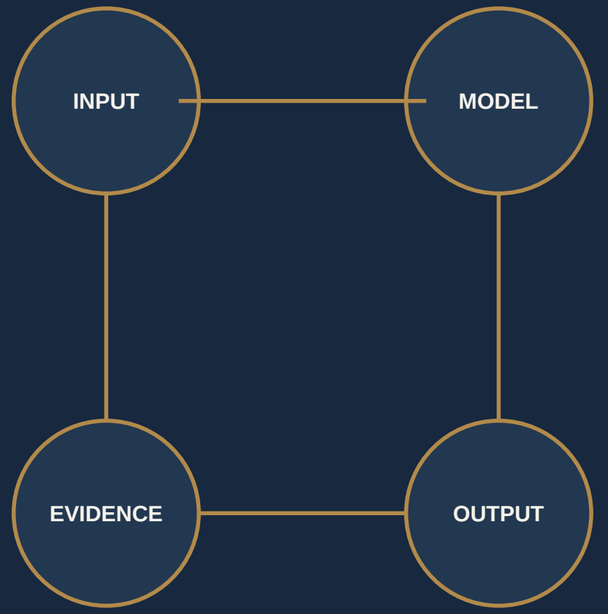
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Swarm Robotics

AI.303 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define swarm intelligence.
- Apply key methods including decentralized control.
- Build or analyze a practical system involving collective behavior.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Swarm intelligence

02

METHODS

Decentralized control

03

SYSTEMS

Collective behavior

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Swarm intelligence

Swarm intelligence

02 · Decentralized control

Decentralized control

03 · Collective behavior

Collective behavior

04 · Communication and distributed

Communication and distributed decision-making

05 · Mandatory practical: Simulate

Mandatory practical: Simulate multiple robots performing coordinated tasks

Core ideas, made explicit

AI.303 • Distinguish definitions, assumptions and evidence

- Swarm intelligence
- Decentralized control
- Collective behavior
- Communication and distributed decision-making
- Mandatory practical: Simulate multiple robots performing coordinated tasks

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.303 • Distinguish definitions, assumptions and evidence

- Swarm intelligence
- Decentralized control
- Collective behavior
- Communication and distributed decision-making
- Mandatory practical: Simulate multiple robots performing coordinated tasks

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Swarm intelligence

INPUTS
Decentralized control

METHOD
Collective behavior

OUTPUT
Communication and distributed decision-making

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving swarm intelligence.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using swarm intelligence.
- Implement or simulate a baseline using decentralized control.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.